

Reality Drift Diagnostic for Silent Drift in LLM Workflows

Detecting Semantic Fidelity Loss When Outputs Still Appear Correct

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Purpose

This diagnostic evaluates LLM workflows that continue producing fluent, useful-looking outputs while becoming less faithful to user intent, source context, task purpose, or external conditions. Silent drift names the observable pattern. It does not assume that the underlying model is the sole source of change.

The unit of evaluation is the complete workflow. Model versions, system instructions, retrieval sources, memory, context compression, tools, agent handoffs, evaluators, users, and downstream reuse can all alter the result. The purpose of the diagnostic is to locate the transformation where fidelity is lost and determine whether existing evaluation can detect it.

What Makes Drift Silent

Traditional monitoring looks for declining accuracy, changing input distributions, increased error rates, or failed tasks. LLM workflows can degrade without producing those signals. Outputs remain coherent, structurally complete, and superficially correct. The first losses often appear in specificity, constraint preservation, contextual fit, uncertainty, provenance, and practical usefulness.

Silent drift exists when the workflow preserves the appearance of successful completion after the relationship between the output and its intended reference has weakened.

Possible Sources of Change

Model

Version change, routing change, decoding settings, safety behavior, latent capability change.

Instructions

Prompt edits, system policy changes, conflicting requirements, template accumulation.

Retrieval

Source replacement, index staleness, ranking changes, missing documents, derivative sources.

Context and memory

Truncation, summarization, stale memory, compressed history, inherited assumptions.

Tools and agents

Handoff loss, tool failure, hidden transformations, recursive generation.

Evaluation

Changed rubrics, synthetic judges, proxy scores, reviewer adaptation.

Users and workflow

Changed prompting habits, automation bias, reduced review, dependency.

External environment

Changed facts, policies, language, user needs, or operating conditions.

Seven Diagnostic Dimensions

1. Intent Preservation

Does the output answer the actual request, or does it merely reproduce the request's surface form?

2. Context Preservation

Do relevant qualifications, hierarchy, tone, relationships, and implied constraints survive?

3. Referential Grounding

Can important claims be traced to source material, observation, or external conditions?

4. Constraint Preservation

Are explicit requirements, exclusions, boundaries, and stopping conditions retained across the workflow?

5. Transformation Depth

How many retrievals, summaries, generations, evaluations, or agent handoffs separate the output from the source?

6. Correction Burden

How much human reconstruction, clarification, verification, or rewriting is required before the output can be used?

7. Coherence-Contact Divergence

Did fluency, confidence, or apparent completeness increase while evidentiary contact weakened?

Transformation Chain Test

For source-based work, preserve and compare each stage:

Source → Retrieved Passage → Model Interpretation → Draft Output → Final Reused Output

At every transition, record:

- What was removed?
- What was added?
- What became more certain?
- Which distinctions were flattened?
- What lost provenance?
- What became harder to invalidate?
- Did a derivative representation return as if it were an independent source?

Longitudinal Evaluation Packet

Repeatedly testing one prompt is not enough. Stochastic variation can look like drift, while a stable answer can conceal a stable defect. Maintain a fixed packet containing:

- Stable prompts and source materials
- Explicit expected constraints and prohibited substitutions
- Known ambiguity and failure traps
- Scoring anchors with examples
- Multiple samples per condition
- Model, system, prompt, retrieval, and tool versions
- Independent grounding evidence
- A record of reviewer judgments and disagreements

Scoring Method

Score each of the seven diagnostic dimensions from 0 to 3.

0 — Preserved

No material loss for the intended use.

1 — Minor loss

Noticeable but recoverable without changing the conclusion or action.

2 — Material loss

Meaning, context, or constraints have changed enough to affect use.

3 — Operationally misleading

The output appears usable but would direct interpretation or action away from the source or purpose.

After scoring, mark every defect that is visible, recoverable, provenance-obscured, recursively amplified, or capable of passing the existing evaluation.

A high total score matters, but a single operationally misleading defect can justify escalation when the use is consequential.

Behavioral Comparison

Do not treat consistency as alignment. A workflow can become consistently generic, consistently overconfident, or consistently wrong. Compare outputs for preservation of purpose and distinctions, not merely similarity. Test paraphrases, reordered constraints, adversarially similar cases, edge cases, and multi-step tasks. Examine whether small early losses compound across later steps.

Provenance and Recursive Reuse

Identify whether generated summaries, evaluations, or answers are being reused as inputs. Recursive Compression allows early losses to persist while later processing makes the result more fluent. Citation presence is insufficient when citations point to derivative material, fail to support the claim, or cannot reconstruct the transformation chain.

Human Review Requirements

- Reviewers must have access to the original source and intended use.
- Reviewers must distinguish factual accuracy from Semantic Fidelity.
- Reviewers must be able to record uncertainty and disagreement.
- Review findings must reach someone authorized to alter the workflow.
- Serious failures must be capable of triggering restriction, rollback, or pause.

Diagnostic Record

Use the following record to document the first material failure and the conditions surrounding it.

Workflow and version: Record the model, prompt, retrieval configuration, memory state, tools, and evaluators in use.

Ground condition: State the task, decision, source, or real-world outcome the workflow is supposed to support.

Failure location: Identify the point in the transformation chain where the first material loss appears.

Affected dimension: Name the diagnostic dimension or dimensions involved.

Severity: Assign a score from 0 to 3.

Existing detection: State whether the current monitoring or evaluation process would have surfaced the defect.

Downstream effect: Describe how the defect changes interpretation, judgment, action, or future inputs.

Corrective action: Specify the required response: reanchor, revise, restrict, roll back, pause, or retire.

Decision Rule

Escalate when an output remains fluent and operationally acceptable while materially losing intent, context, grounding, constraints, or provenance.

The decisive question is not whether the output looks correct. It is whether the workflow remains answerable to what the output is supposed to represent.

Coherence is not contact.